LLM Environment Open Data: an Open Dataset, API, and MCP Server for the Quantitative Literature on the Environmental Footprint of Large Language Models

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Abstract

The literature on the environmental footprint of large language models now provides a growing number of quantified indicators, but these results remain scattered across journal articles, model cards, technical reports, and institutional syntheses. This dispersion slows comparison across studies and limits cumulative reuse. We present *LLM Environment Open Data*, a structured dataset that extracts, normalizes, and documents quantitative values reported on LLM training, inference, infrastructure, and lifecycle impacts. The project comprises four complementary artifacts: an open tabular dataset, a layer of comparability metadata, a local HTTP API, and an MCP server for agent-oriented querying. The current release contains 76 records spanning 11 sources, and links each value to its unit, system boundary, analytical scope, and precise source locator. We further show how this evidence base can support an inference-only screening estimator for current LLM services, including proprietary models that do not disclose direct environmental telemetry. The estimator remains explicitly extrapolative: it starts from observed literature anchors in native units (`Wh/prompt|query` and `Wh/page`), scales them by model size, and derives carbon and water through country electricity mixes. This paper describes the dataset construction process, the estimation method, its limitations, and its value for both reproducible quantitative reviews of AI environmental impacts and environmentally responsible software design.

Keywords: LLMs, open data, dataset, environmental footprint, API, MCP, quantitative review.

1 Introduction

Debate on the environmental footprint of LLMs has intensified as publications have multiplied on training, inference, water use, data centers, and hardware lifecycle effects [15, 13, 9, 12, 11, 5]. Yet a simple methodological problem remains: quantitative values are distributed across heterogeneous formats, with highly variable units of analysis, system boundaries, and documentary locators. This heterogeneity makes cumulative reuse difficult.

The aim of this paper is to document the data infrastructure associated with our quantitative review of LLMs and the environment. The primary contribution is an open dataset that stores extracted values together with comparability metadata and machine-access interfaces through an API and an MCP server. This infrastructure also serves as the basis for a first inference estimator for practical use cases, such as estimating the annualized inference footprint of a software feature or comparing current LLM services under a common usage convention.

2 Positioning

The existing literature already provides robust anchor points, for example the training emissions of BLOOM [9], the GPU-hours and emissions documented by Meta for Llama 3.1 [10], production-scale inference measurements at Google [5], water-footprint estimates [8], and annual electricity trajectories for data centers [6, 14, 7, 4]. However, the literature provides few open infrastructures designed to consolidate such results into a directly reusable form.

The *LLM Environment Open Data* project addresses this gap through three simple design choices:

- one record per quantified value;
- explicit traceability to the page, table, section, or paragraph of origin;
- exposure of the data for both human readers and software agents.

3 Dataset Construction Method

The dataset is built from the quantitative literature review associated with the main publication on LLMs and the environment. For each retained source, we create one record for each quantified value that can be directly reused in the review. A single study may therefore generate several records when it reports multiple metrics or multiple units of analysis.

Each record contains the following fields:

- unique record identifier;
- source key;
- short citation;
- publication year;
- analytical phase: `training`, `inference`, `infrastructure`, `lifecycle`;
- impact category: carbon, energy, water, or compute;
- metric name;
- value and unit;
- model or scope concerned;
- geography;
- temporal scope;
- system boundary;
- data type: measured, calculated, modeled, projected, estimated, or statistical;
- source locator;
- source URL;
- free-text note.

This structure makes it possible to separate three dimensions that are often conflated in narrative reviews: the observed phenomenon, the way it was measured, and the exact level to which it applies.

4 Initial Release Content

The current release contains 76 records spanning 11 sources. All four analytical phases are represented: 14 records for training, 17 for inference, 17 for infrastructure, and 28 for lifecycle effects. Energy and carbon are the most represented impact categories, followed by water.

Source	Phase	Example metric	Example value
Strubell et al. (2019) [15]	training	training emissions	626 155 lb CO_2e
Luccioni et al. (2023) [9]	training	BLOOM 176B emissions	24.7–50.5 tCO_2e
Meta (2024) [10]	training	Llama 3.1 405B GPU-hours	30.84 million GPU-hours
Elsworth et al. (2025) [5]	inference	energy per prompt	0.24 Wh/prompt
Ren et al. (2024) [12]	inference	energy per 500-word page	0.0195 kWh/page
Li et al. (2025) [8]	lifecycle / water	GPT-3 total training water use	5.4 million liters
IEA (2025) [6]	infrastructure	annual global data center electricity	415 TWh in 2024
Berkeley Lab (2025) [7]	infrastructure	annual US data center electricity	176 TWh in 2023
de Vries-Gao (2025) [4]	infrastructure	annualized global AI power demand	46.4–201.5 TWh/year derived
Morrison et al. (2025) [11]	lifecycle	total water over model creation pipeline	2.769 million liters

Table 1: Representative records from the initial dataset release.

5 Formats and Schema

The dataset is distributed in two synchronized forms: a canonical CSV file and a JSON export. The CSV facilitates human auditing, spreadsheet-based review, and version control. The JSON format facilitates programmatic integration into scripts, agents, or web interfaces. A JSON schema documents the expected structure and defines required fields.

The decision to maintain a canonical tabular source responds to a practical constraint: bibliographic extractions are often first reviewed and corrected manually. JSON is then regenerated from the CSV. This separation avoids multiplying sources of truth.

The main dataset is complemented by a second metadata table keyed by `record_id`. This layer makes explicit estimation-oriented fields such as `unit_of_analysis`, `source_type`, `uncertainty_level`, `raw_or_derived`, and `applicability_domain`. The methodological point is important: these fields are no longer only reconstructed in code, but published as data and therefore remain auditable, editable, and versionable.

6 Local HTTP API

The project provides a local HTTP API for rapid exploration of the dataset. The main endpoints are:

- `/health` to verify service status;
- `/records` to list records with simple filters;
- `/records/<record_id>` to retrieve a specific record;
- `/sources` to list distinct source studies;
- `/stats` to retrieve summary statistics;
- `/estimate` to produce request-level estimates;
- `/estimate_feature` to produce annualized feature-level inference estimates;
- `/predict_inference` to expose the inference predictor directly;
- `/models`, `/market-models`, and `/energy-mix` to expose the model and country catalogs used by the estimator.

This API is not intended to replace a full open-data platform. Rather, it provides a minimal programmatic reuse layer directly associated with the manuscript and the data repository [2]. The estimation endpoints extend the role of the dataset: it is no longer only an archive of reported values, but also a support for traceable calculations grounded in literature-derived factors.

7 MCP Server

The project also provides an MCP server over `stdio`. Its purpose differs from that of the API: it enables software agents to query the dataset directly as structured context, without scraping or heuristic reading of tables in the manuscript.

The current version exposes thirteen tools:

- `list_records`;
- `get_record`;
- `aggregate_by_phase`;
- `list_sources`;
- `list_models`;
- `list_market_models`;
- `get_model_profile`;
- `list_country_energy_mix`;
- `get_country_energy_mix`;
- `list_extrapolation_rules`;
- `predict_inference_externalities`;
- `estimate_externalities`;

- `estimate_feature_externalities`.

This design addresses a precise need: enabling an assistant or software tool to recover a source-backed quantitative value, verify its analytical phase, and trace it back to the underlying publication without returning to an unrestricted reading of the PDF [3].

8 From Data to Estimation

One of the central use cases of the project is the following question: "Here is my application and the LLM it uses. Can you estimate the environmental externalities of inference?" The central difficulty is that providers of proprietary services rarely publish direct telemetry. The estimator therefore does not claim to measure hidden provider data. Instead, it constructs a transparent extrapolation from the inference indicators that are actually available in the literature.

The current estimator is deliberately restricted to *inference*. Training, embodied impacts, and application-side software overheads are not included in the displayed result. This restriction is methodological: the available inference indicators are scarce but still sufficiently structured to support a screening model, whereas mixing them with training or infrastructure indicators would blur system boundaries.

The current implementation follows four rules.

- It starts from *observed energy anchors* reported in the literature, not from provider claims. In the current release, the predictive core relies on one `Wh/prompt` anchor from Elsworth et al. [5] and two `kWh/page` anchors from Ren et al. [12].
- It keeps `prompt|query` and `page` as *separate native unit families*. The project does not currently infer a universal prompt-to-page conversion because the corpus does not yet contain a sufficiently documented same-model bridge.
- It uses model size as the extrapolation variable. More precisely, for each family of indicators, the estimator computes an average energetic intensity per billion active parameters and applies it to the target model.
- It derives carbon and water from the retained energy estimate using country-level electricity intensities. Country factors are therefore contextual parameters, not model measurements.

The project therefore makes an explicit distinction between three types of values. First, literature anchors such as "0.24 Wh/prompt" for Gemini Apps [5] or "0.0195 kWh/page" for Llama 3 70B [12] remain observed values. Second, model-size estimates for current proprietary services are sometimes estimated rather than published, and are labeled as such in the catalog. Third, the final inference estimate returned by the API, the MCP server, or the web interface is a derived value obtained by combining literature anchors, model-size scaling, country electricity contextualization, and annualization.

8.1 Final Calculation Equations

Let P_t denote the active parameter count, in billions, of the target model. The predictive core is built from energy anchors only. For each anchor a, let P_a denote the source model size and let E_a denote the published energy factor converted into Wh in its native inference family.

The current implementation separates two anchor families:

- A_{pq} for `Wh/prompt|query`;
- A_{pg} for `Wh/page`.

No empirical universal conversion between these two families is currently assumed.

For each family $f \in \{pq, pg\}$, the estimator computes the mean energetic intensity per billion parameters:

$$I_f = \frac{1}{|A_f|} \sum_{a \in A_f} \frac{E_a}{P_a}. \tag{1}$$

The family-specific per-request or per-page energy estimate for the target model is then:

$$\hat{E}_{t,f} = I_f \times P_t. \tag{2}$$

This is a linear screening extrapolation on model size. It is not presented as a physical law of inference scaling, but as a transparent multiple-based approximation grounded in the few observed anchors currently available.

Carbon and water are then derived from the electricity mix of the retained country. Let CI_c be the carbon intensity of country c in gCO_2e/kWh and WI_c its water intensity in L/kWh. Starting from $\hat{E}_{t,f}$ expressed in Wh:

$$\hat{C}_{t,f,c} = \frac{\hat{E}_{t,f}}{1000} \times CI_c, \tag{3}$$

$$\hat{W}_{t,f,c} = \frac{\hat{E}_{t,f}}{1000} \times WI_c \times 1000. \tag{4}$$

The country c itself depends on the deployment assumption. In the current web application, proprietary hosted services are contextualized with the publisher-country mix retained in the catalog, while open-weight models are contextualized with the project country when it is specified by the user.

At the software-feature level, let n_f be the number of LLM requests per feature use, u_m the monthly number of feature uses, and m_y the number of months per year. Annualized feature uses and annualized LLM requests are:

$$U_{year} = u_m \times m_y, \tag{5}$$

$$N_{LLM,year} = U_{year} \times n_f. \tag{6}$$

The annual direct LLM inference impact for each family is then:

$$E_{LLM,year,f} = \hat{E}_{t,f} \times N_{LLM,year}, \tag{7}$$

$$C_{LLM,year,f} = \hat{C}_{t,f,c} \times N_{LLM,year}, \tag{8}$$

$$W_{LLM,year,f} = \hat{W}_{t,f,c} \times N_{LLM,year}. \tag{9}$$

The important methodological point is that the project reports the prompt/query-based and page-based estimates *separately*. At the current stage of the corpus, collapsing them into a single final figure would create an illusion of precision that the literature does not support.

9 Research Questions

The transition from evidence base to operational estimation leads to four research questions:

- Which metadata are required to make published environmental factors genuinely comparable and reusable?
- To what extent can LLM externalities be estimated from the literature when direct provider telemetry is unavailable?
- How should LLM calls be integrated into a broader software environmental accounting boundary?
- How does machine-access exposure through an API and MCP improve practical reuse for environmentally responsible AI and software design?

10 Implementation Example: From Prompt to Software Feature

The current release includes a first-level estimator designed for screening use. When the target model is present in the model catalog, or when the user provides an estimated active parameter count, the estimator computes two inference estimates in parallel: one derived from the mean `Wh/prompt|query` intensity per billion parameters, and one derived from the mean `Wh/page` intensity per billion parameters.

Consider a demonstration scenario corresponding to GPT-4o-mini, 1 200 input tokens, 350 output tokens, one LLM request per feature use, 10 000 feature uses per month, and a French project context. In the current catalog, GPT-4o-mini is associated with an approximate 8B active-parameter profile and is treated as a proprietary hosted service. The country retained for contextualization is therefore the publisher country rather than France itself.

In the current corpus, the prompt/query family is anchored by the Gemini Apps production measurement reported by Elsworth et al. [5], while the page family is anchored by the Gemma-2B-it and Llama 3 70B measurements reported by Ren et al. [12]. For this reason, the API, the MCP server, and the web interface return two distinct annualized inference estimates instead of one merged figure. The prompt/query estimate remains close to the operational prompt-serving measurement, whereas the page estimate reflects a different measurement family and therefore a different order of magnitude.

This implementation is deliberately conservative. It does not claim to provide an auditable environmental assessment in the sense of formal carbon accounting. Rather, it provides a traceable, reproducible, and extensible estimate that is useful for eco-design, service comparison, and the progressive construction of a broader evidence base on inference impacts.

11 Evaluation Across Three Software Scenarios

To assess the usefulness of the proposed framework, we applied the estimator to three representative software scenarios: a support chatbot, a retrieval-augmented assistant, and a batch-generation workflow. All scenarios rely on the same inference-only estimation logic, but differ in token volume, number of LLM calls per use, and monthly usage volume. Across these scenarios, the method exhibits three stable properties. First, moving from unit-level usage to annualized software features substantially changes the relevant order of magnitude for decision-making. Second, the choice of target model profile and country mix has a material effect on the result even when the user-facing task remains the same. Third, the prompt/query and page families should still be interpreted as parallel screening estimates rather than as interchangeable measurements of one single latent quantity. For that reason, the current web application and model catalog keep both families visible rather than collapsing them into one allegedly definitive value.

12 Scientific Contribution

The main contribution is not algorithmic but infrastructural. The publication associated with the dataset transforms a literature review into a reusable evidence base. This changes the nature of the scientific deliverable in four ways.

First, it separates the narrative layer of the paper from the reusable factual layer. Second, it enables incremental updates of the extraction base without rewriting the entire synthesis. Third, it provides a common support for comparing results that belong to different analytical levels, such as a query, a prompt, a generated page, a final training run, or an annual data center electricity figure. Fourth, it serves as an operational basis for an externalities estimator for software systems integrating LLMs, thereby bringing the quantitative review closer to the concerns of responsible digital design and software environmental accounting.

13 Limitations

The dataset does not eliminate the underlying heterogeneity of the literature; it makes that heterogeneity explicit. Values remain dependent on assumptions, calculation methods, and system boundaries. Some sources are peer-reviewed papers, while others are model cards or technical reports. The dataset therefore includes metadata on evidence type and scope precisely to reduce invalid comparisons.

The current release is also partial. It primarily covers the values mobilized in the associated review paper and the major sources identified so far. The objective is not to claim immediate exhaustiveness, but to provide an open infrastructure that can be enriched, audited, and corrected over time.

14 Conclusion

This paper accompanies the release of *LLM Environment Open Data*, a structured dataset on quantified environmental impacts of LLMs and their infrastructure. By combining an open dataset, a layer of comparability metadata, an HTTP API, and an MCP server, it proposes an augmented form of scientific publication: readers receive not only a synthesis, but also the evidence layer that makes that synthesis verifiable, extensible, and automatable [1, 2, 3].

The next logical step is to expand the source corpus and publish the dataset with a persistent identifier. At this stage, the project already provides the necessary scaffolding to treat the dataset as a scientific object in its own right, while also serving as the starting point for a method of environmental accounting for software systems integrating LLMs.

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